Vectors I Basics Worksheet: Solutions

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Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points and the ratio theorem. Each section has footing questions, one tricky question and a True/False check.

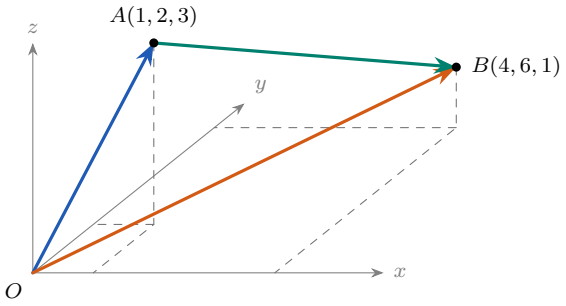
1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ as a column vector, and write $\mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$ in terms of **i**, **j** and **k**.

$$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$$

$$\begin{aligned} \mathbf{b} &= 4\mathbf{i} + 0\mathbf{j} - 1\mathbf{k} \\ &= 4\mathbf{i} - \mathbf{k} \end{aligned}$$

2 The points A and B have coordinates $(1, 2, 3)$ and $(4, 6, 1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{pmatrix} 4 \\ 6 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} \\ &= 3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k} \end{aligned}$$

3 (Tricky) Given $\mathbf{p} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{q} = -\mathbf{i} + 3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p} - 3\mathbf{q} + \mathbf{r} = \mathbf{0}$. Give $\mathbf{r}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

$$\mathbf{p} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$$

$$\begin{aligned} \mathbf{r} &= 3\mathbf{q} - 2\mathbf{p} \\ &= 3\begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -3 \\ 9 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} -7 \\ 9 \\ -2 \end{pmatrix} \\ &= -7\mathbf{i} + 9\mathbf{j} - 2\mathbf{k} \end{aligned}$$

Check Your Understanding.

	Statement	True / False
(a)	$3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$	**False**
(b)	The $\mathbf{j}$-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1	**True**
(c)	The position vector of $P(1,-2,0)$ is $\mathbf{i}-2\mathbf{j}$	**True**
(d)	$\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$	**False**
(e)	$\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$	**True**

(a) $3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\0\\2\end{pmatrix}\neq\begin{pmatrix}3\\2\\0\end{pmatrix}$

(b) $4\mathbf{i}-\mathbf{j}+2\mathbf{k}=\begin{pmatrix}4\\-1\\2\end{pmatrix}$

(c) $\overrightarrow{OP}=\begin{pmatrix}1\\-2\\0\end{pmatrix}=\mathbf{i}-2\mathbf{j}$

(d) $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$

(e) $\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$

2 Magnitude

1 Find $|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}|$.

$$\begin{aligned}|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}| &= \sqrt{2^2+(-3)^2+6^2}\\ &= \sqrt{49}\\ &= 7\end{aligned}$$

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix} \\ AB = \left|\overrightarrow{AB}\right| &= \sqrt{2^2 + (-6)^2 + 3^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

3 **(Tricky)** Given $\mathbf{a} = \begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}| = 3$, find the possible values of t.

$$\begin{aligned}|\mathbf{a}|^2 &= t^2 + 2^2 + (t-1)^2 \\ 9 &= t^2 + 4 + t^2 - 2t + 1 \\ 0 &= 2t^2 - 2t - 4 \\ 0 &= t^2 - t - 2 \\ 0 &= (t-2)(t+1) \\ t &= 2 \text{ or } t = -1\end{aligned}$$

Check Your Understanding.

	Statement	**True / False**
(a)	$\lvert\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}\rvert = 1 + 2 - 2 = 1$	**False**
(b)	$\lvert -\mathbf{a}\rvert = \lvert\mathbf{a}\rvert$ for every vector $\mathbf{a}$	**True**
(c)	$\lvert\mathbf{a} + \mathbf{b}\rvert = \lvert\mathbf{a}\rvert + \lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$	**False**
(d)	$\lvert 3\mathbf{a}\rvert = 3\lvert\mathbf{a}\rvert$	**True**
(e)	$\left\lvert \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix} \right\rvert = -5$	**False**
(f)	$\left\lvert \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\rvert = 1$	**False**
(g)	$\lvert\mathbf{a} + \mathbf{b}\rvert \leq \lvert\mathbf{a}\rvert + \lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$	**True**

(a) $|\mathbf{i}+2\mathbf{j}-2\mathbf{k}| = \sqrt{1^2+2^2+(-2)^2} = 3$

(b) $|-\mathbf{a}| = |-1|\,|\mathbf{a}| = |\mathbf{a}|$

(c) $|\mathbf{i}+\mathbf{j}| = \sqrt{2} \neq |\mathbf{i}|+|\mathbf{j}| = 2$

(d) $|3\mathbf{a}| = |3|\,|\mathbf{a}| = 3|\mathbf{a}|$

(e) $\left|\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right| = \sqrt{0^2+(-5)^2+0^2} = 5$

(f) $\left|\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right| = \sqrt{1^2+1^2+1^2} = \sqrt{3} \neq 1$

(g) $\overrightarrow{OA} = \mathbf{a},\ \overrightarrow{AB} = \mathbf{b},\ \overrightarrow{OB} = \mathbf{a}+\mathbf{b},\ OB \leq OA + AB\ (\triangle OAB)$

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

$$\left|\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}\right| = \sqrt{2^2+(-1)^2+2^2}$$

$$= 3$$

$$\text{unit vector} = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

$$\left| \begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix} \right| = \sqrt{3^2 + 0^2 + (-4)^2}$$
$$= 5$$
$$\text{vector} = 10 \cdot \frac{1}{5} \begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$$
$$= \begin{pmatrix} 6 \\ 0 \\ -8 \end{pmatrix}$$

3 **(Tricky)** The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$\left| \overrightarrow{AB} \right| = \sqrt{2^2 + (-6)^2 + 3^2}$$
$$= 7$$
$$\text{vector} = -14 \cdot \frac{1}{7} \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} -4 \\ 12 \\ -6 \end{pmatrix}$$

Check Your Understanding.

	Statement	True / False
(a)	$\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector	**False**
(b)	$\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector	**True**
(c)	$\mathbf{i}+\mathbf{j}$ is a unit vector	**False**
(d)	$\lvert\hat{\mathbf{a}}\rvert = 1$ for every non-zero vector $\mathbf{a}$	**True**
(e)	$\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel	**True**

(a) $\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right| = \sqrt{3} \neq 1$

(b) $\left|\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}\right| = \frac{1}{3}\sqrt{1+4+4} = 1$

(c) $|\mathbf{i}+\mathbf{j}| = \sqrt{1^2+1^2+0^2} = \sqrt{2} \neq 1$

(d) $|\hat{\mathbf{a}}| = \left|\frac{\mathbf{a}}{|\mathbf{a}|}\right| = \frac{|\mathbf{a}|}{|\mathbf{a}|} = 1$

(e) $\mathbf{a} = |\mathbf{a}|\,\hat{\mathbf{a}}$

4 Parallel Vectors ($\mathbf{a} = \lambda\mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}-6\\3\\-9\end{pmatrix}$ are parallel.

$$\mathbf{b} = \begin{pmatrix}-6\\3\\-9\end{pmatrix}$$
$$= -3\begin{pmatrix}2\\-1\\3\end{pmatrix}$$
$$= -3\mathbf{a}$$
$$\therefore\ \mathbf{a} \parallel \mathbf{b} \text{ (shown)}$$

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

$$\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix} = \lambda \begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$$
$$4 = -2\lambda$$
$$\lambda = -2$$
$$p = 3\lambda$$
$$= -6$$
$$-2 = \lambda q$$
$$-2 = -2q$$
$$q = 1$$

3 **(Tricky)** Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

$$|\mathbf{a}| = \sqrt{1^2 + 2^2 + (-2)^2}$$
$$= 3$$
$$\text{vector} = \pm 6 \cdot \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$
$$= \pm \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$$

Check Your Understanding.

	Statement	True / False
(a)	If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$	**False**
(b)	$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$	**True**
(c)	$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$	**False**
(d)	$\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$	**True**
(e)	If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions	**False**

(a) $2\mathbf{i} \parallel \mathbf{i},\ 2\mathbf{i} \neq \mathbf{i}$

(b) $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$

(c) $\frac{2}{1} = \frac{4}{2} \neq \frac{6}{4}$

(d) $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix} = -3 \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$

(e) $\mathbf{a} = \lambda\mathbf{b},\ \lambda = -2$

5 Collinear Points

1 Show that the points $A(1,2,3)$, $B(3,5,4)$ and $C(7,11,6)$ are collinear.

$$\begin{aligned}
\overrightarrow{AB} &= \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\
&= \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \\
\overrightarrow{BC} &= \begin{pmatrix} 7 \\ 11 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} \\
&= \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix} \\
&= 2\overrightarrow{AB} \\
\therefore\ \overrightarrow{AB} &\parallel \overrightarrow{BC},\ B \text{ common} \\
\therefore\ A, B, C &\text{ collinear (shown)}
\end{aligned}$$

2 For the points in Question 1, find the ratio $AB : BC$.

$$\begin{aligned}
\overrightarrow{BC} &= 2\overrightarrow{AB} \\
BC &= 2AB \\
AB : BC &= 1 : 2
\end{aligned}$$

3 **(Tricky)** The points $P(2,-1,0)$, $Q(4,1,3)$ and $R(a,7,b)$ are collinear. Find a and b.

$$\begin{aligned}
\overrightarrow{PR} &= \lambda\overrightarrow{PQ} \\
\begin{pmatrix} a-2 \\ 8 \\ b \end{pmatrix} &= \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \\
8 &= 2\lambda \\
\lambda &= 4 \\
a - 2 &= 2(4) \\
a &= 10 \\
b &= 3(4) \\
&= 12
\end{aligned}$$

Check Your Understanding.

	Statement	True / False
(a)	If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear	**False**
(b)	If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear	**True**
(c)	$A(0,0,0)$, $B(1,1,1)$ and $C(2,2,3)$ are collinear	**False**
(d)	The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$)	**True**

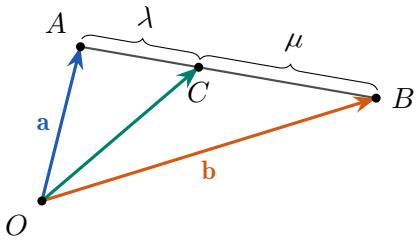
(a) $A(0,0,0), B(1,0,0), C(0,1,0), D(1,1,0)$: $\overrightarrow{AB} = \overrightarrow{CD}$

(b) $\overrightarrow{AB} \parallel \overrightarrow{AC}$, A common

(c) $\overrightarrow{AC} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \neq \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

(d) $2\mathbf{a} - \mathbf{a} = \mathbf{a}$, $-\mathbf{a} - \mathbf{a} = -2(\mathbf{a})$

6 Ratio Theorem



In the True/False table, C lies on AB with $AC : CB = \lambda : \mu$, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

1 Let $\overrightarrow{OA} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\overrightarrow{OB} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

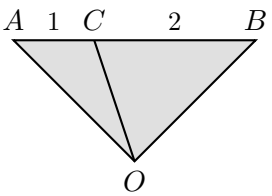
(a) $AC : CB = 1 : 2$,

(b) $3AC = 2CB$,

(c) $2AB = 5CB$,

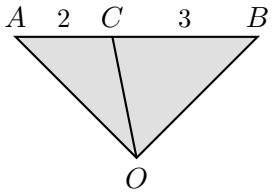
(d) Given instead that C lies on AB produced, and $3AC = 7BC$.

(a)



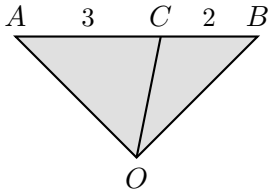
$$\overrightarrow{OC} = \frac{2\overrightarrow{OA} + \overrightarrow{OB}}{3}$$
$$= \begin{pmatrix} 1 \\ \frac{4}{3} \\ \frac{5}{3} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{4}{3}, \tfrac{5}{3}\right)$$

(b)



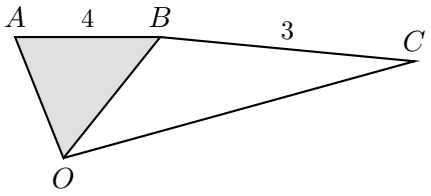
$$\frac{AC}{CB} = \frac{2}{3}$$
$$\overrightarrow{OC} = \frac{2\overrightarrow{OB} + 3\overrightarrow{OA}}{5}$$
$$= \begin{pmatrix} 1 \\ \frac{7}{5} \\ \frac{9}{5} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{7}{5}, \tfrac{9}{5}\right)$$

(c)



$$\frac{AB}{CB} = \frac{5}{2}$$
$$\overrightarrow{OC} = \frac{3\overrightarrow{OB} + 2\overrightarrow{OA}}{5}$$
$$= \begin{pmatrix} 1 \\ \frac{8}{5} \\ \frac{11}{5} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{8}{5}, \tfrac{11}{5}\right)$$

(d)



$$\begin{aligned}
\frac{AC}{BC} &= \frac{7}{3} \\
\overrightarrow{OB} &= \frac{3\overrightarrow{OA} + 4\overrightarrow{OC}}{7} \\
\overrightarrow{OC} &= \frac{7\overrightarrow{OB} - 3\overrightarrow{OA}}{4} \\
&= \begin{pmatrix} 1 \\ \frac{11}{4} \\ \frac{9}{2} \end{pmatrix} \\
\therefore\ C&\left(1, \tfrac{11}{4}, \tfrac{9}{2}\right)
\end{aligned}$$

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3, -1, 5)$ and $(-1, 5, 1)$.

$$\begin{aligned}
\overrightarrow{OM} &= \frac{1}{2}\left[\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix} + \begin{pmatrix} -1 \\ 5 \\ 1 \end{pmatrix}\right] \\
&= \frac{1}{2}\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \\
&= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}
\end{aligned}$$

Check Your Understanding.

	Statement	**True / False**
(a)	$\overrightarrow{OC} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$	**True**
(b)	$\overrightarrow{OC} = \dfrac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$	**False**
(c)	If $\lambda : \mu = 1 : 1$, then $\overrightarrow{OC} = \dfrac{1}{2}(\mathbf{a} + \mathbf{b})$	**True**
(d)	If $\lambda : \mu = 1 : 3$, then $\overrightarrow{OC} = \dfrac{3\mathbf{a} + \mathbf{b}}{4}$	**True**
(e)	If $\lambda : \mu = 2 : 1$, then $\overrightarrow{OC} = \dfrac{2\mathbf{a} + \mathbf{b}}{3}$	**False**

(a) $\overrightarrow{OC} = \mathbf{a} + \frac{\lambda}{\lambda+\mu}(\mathbf{b}-\mathbf{a}) = \frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(b) $\overrightarrow{OC} = \frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(c) $\overrightarrow{OC} = \frac{1\mathbf{a}+1\mathbf{b}}{2}$

(d) $\overrightarrow{OC} = \frac{3\mathbf{a}+1\mathbf{b}}{1+3}$

(e) $\overrightarrow{OC} = \frac{1\mathbf{a}+2\mathbf{b}}{3}$

END OF WORKSHEET